

Tanvir Singh

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Education

Macquarie University

Bachelor of Information Technology – Software Technology Major

North Ryde, NSW

Expected Dec 2026

• **WAM:** 75+

• **Relevant Coursework:** Algorithms & Data Structures, Object-Oriented Programming, Systems Programming, Distributed Systems, Mobile Application Development, Database Design, Computer Networks

Projects

QuestShaper Multiplayer Game Server

github.com/TanvirS-07/QuestShaper

Java, REST APIs, Docker, AWS, Terraform

- Led backend development within a 3-person team, implementing the majority of the Java REST API for authentication, game state, inventory, and player actions in a multiplayer game server.
- Independently took care of the cloud infrastructure and CI/CD: containerised the application with Docker, automated testing and deployment via GitHub Actions, and provisioned AWS EC2 infrastructure as code using Terraform.

Distributed Systems Job Scheduler

github.com/TanvirS-07/Job-Scheduler

Python, TCP Sockets, XML

- Co-developed (with one partner) a client-server job scheduler for a simulated environment using socket-based communication and XML message handling, reducing average job turnaround time by over 99% versus baseline scheduling algorithms.
- Authored technical documentation with data visualisations analysing performance trade-offs between resource utilisation, cost, and turnaround time across workloads of up to 10,000 jobs.

DormDeals – Campus Marketplace App

github.com/TanvirS-07/DormDeals

Flutter, Dart, Firebase, Google APIs, Git

- Built a cross-platform Android and web marketplace independently, developing the application end-to-end across frontend, backend, authentication, database, and location-based features.
- Implemented Firebase Authentication and a real-time Firestore database to manage user accounts and listings, with Provider-based state management.
- Integrated Google Maps APIs for location-based listing features.

Virtual Museum – Computing Industry Project

github.com/TanvirS-07/Virtual-Museum

Unity, C#, Jira, Git

- Collaborated in a 5-person Agile team to deliver an interactive first-person Unity exhibit for an external client (Macquarie University Library Exhibition).
- Served as developer and documentation lead, implementing gameplay features in C# and authoring project documentation across multiple Scrum sprints, incorporating client and playtester feedback.

Experience

Co-Founder & Technical Operations Lead

March 2025 – Present

Precision Cricket Academy

Schofields, NSW

- Co-founded and operated a cricket training academy, coordinating player schedules, payments, inventory, and day-to-day operations across training programs.
- Engineered a digital system to track player schedules, manage payments, and organise training programs.
- Worked closely with players, parents, and local cricket clubs to understand their needs and provide a positive training experience.

Academic Tutor (Mathematics & STEM)

April 2019 – June 2025

Kumon Education

The Hills District, NSW

- Supported students in mathematics, problem-solving, and core STEM concepts.
- Explained challenging mathematical and logical concepts in clear, step-by-step ways suited to each student's learning pace.
- Monitored student progress and adjusted teaching approaches to address areas of difficulty and support academic improvement.

Technical Skills

Languages: Python, Java, C, C++, C#, Dart, SQL, Bash

Frameworks & APIs: Flutter, Android SDK, REST APIs, Firebase, Node.js, Google APIs

Cloud & DevOps: AWS EC2, Docker, Terraform, GitHub Actions, CI/CD

Databases: Firestore, MySQL

Tools: Git, GitHub, Maven, JUnit, Linux, GDB, Wireshark

Networking: TCP/IP, Routing, Switching, Subnetting, Ethernet

Certifications: AWS Certified Cloud Practitioner, GitHub Foundations, HackerRank Software Engineer, HackerRank SQL (Advanced), HackerRank REST API (Intermediate)